

Master 1 - LAS

Machine learning for IoT

Prediction of Time Series

Organization

- Tuesday, May 26 2026 / Thursday, May 28, 2026 - Supervisor: Michel Salomon
 - **Software labs:** a little warm-up with two Colab files to complete
 - [Lab 1](#) - Logistic Regression and Metrics
 - [Lab 2](#) - Decision Trees - Random Forests
 - Prediction / classification **Task 1** in the context of data collected during [LAS Realisation and development of IoT applications](#) supervised by Joseph Azar
 - You must apply what you have learned in machine learning on some data you collected during the first LAS. You can also investigate / propose new approaches
 - Prediction / classification **Task 2** in the smart city context: see next slides
- **Deadline to send files - Labs and Tasks: Saturday, June 6 2026 - 6PM**
- Wednesday, May 27 2026 - Supervisor: Jacques Bahi
 - Prediction of stock exchange evolution, etc.

Instructions

- On the following slides you will find a short description of the program you have to write for the smart city context. It is a program dealing with the prediction / classification of time series.
- This time series task is based on a paper that was published.
- The program must be a Colab file.
- You can work in groups as you did in the LAS of the previous semester.

All files must be send to michel.salomon@umlp.fr

Prediction / classification - smart city context

- The goal is to study and reproduce some of the results obtained in paper [Deep Learning and Gradient Boosting for Urban Environmental Noise Monitoring in Smart Cities](#)
- You must carefully read the paper and write a brief summary of it
- Implement and train some models: Gradient Boosting and others you choose
 - Data provided by [DataParkmeter.zip](#)
 - Three parameters are considered, resulting in 3 CSV files
 - Plot the curves of the three parameters, but use only Parkmeter 515 for ML
 - Preprocess data as explained in Equation (1)
 - Use MinMaxScaler from Scikit-learn to normalize data
 - You must add comments to explain / present all the steps from the data loading up to the prediction results (use them to detect false data injection attacks as done in the paper)

Prediction / classification - smart city context

- Your program must be composed of the following steps

Step 1 - Basic imports (pandas, numpy, etc.)

Step 2 - Load data from the three parkmeters

Step 3 - Process the 77 histogram columns : calculate the average noise according to Equation (1)

Step 4 - Resample data to 1-hour intervals and filter to the period May 1, 2019 to January 31, 2020

Step 5 - Visualize hourly noise trends like in Figure 2

Step 6 - Normalize data of parkmeter 515 with MinMaxScaler and split in train and test (try ratios)

Prediction / classification - smart city context

- Your program must be composed of the following steps (continued)

Step 7 - Create LightGBM dataset controlled by a look_back value

Step 8 - Train LightGBM according to Config 1 defined by the upper line of Table 2

Step 9 - Train LightGBM according to Config 2 defined by the lower line of Table 2

Step 10 - Calculate LightGBM RMSE (do not forget to inverse the values)

Step 11 - Apply the best configuration to 6-days forecasting as in Figure 4

Step 12 - Investigate alternative ML models and compare them with LightGBM